SLE-3: ARCHITECTURAL DESIGN USING FULL C4 MODEL

Course: 02AML204 – Introduction to Artificial Intelligence

System: Traffic Signal Optimizer Agent using BFS and DFS

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1. System Title & Short Description

Traffic Signal Optimizer Agent is an AI-based search system that models traffic conditions at an intersection as a graph of traffic states. Each state represents a possible signal/traffic condition, and transitions represent possible changes in signal phase. The system uses Breadth-First Search (BFS) and Depth-First Search (DFS), carried forward from SLE-2, to search the state graph and identify a path toward a desired traffic condition. This report describes the architecture using the four levels of the C4 model: Context, Container, Component and Code.

2. Context Diagram (C4 Level 1)

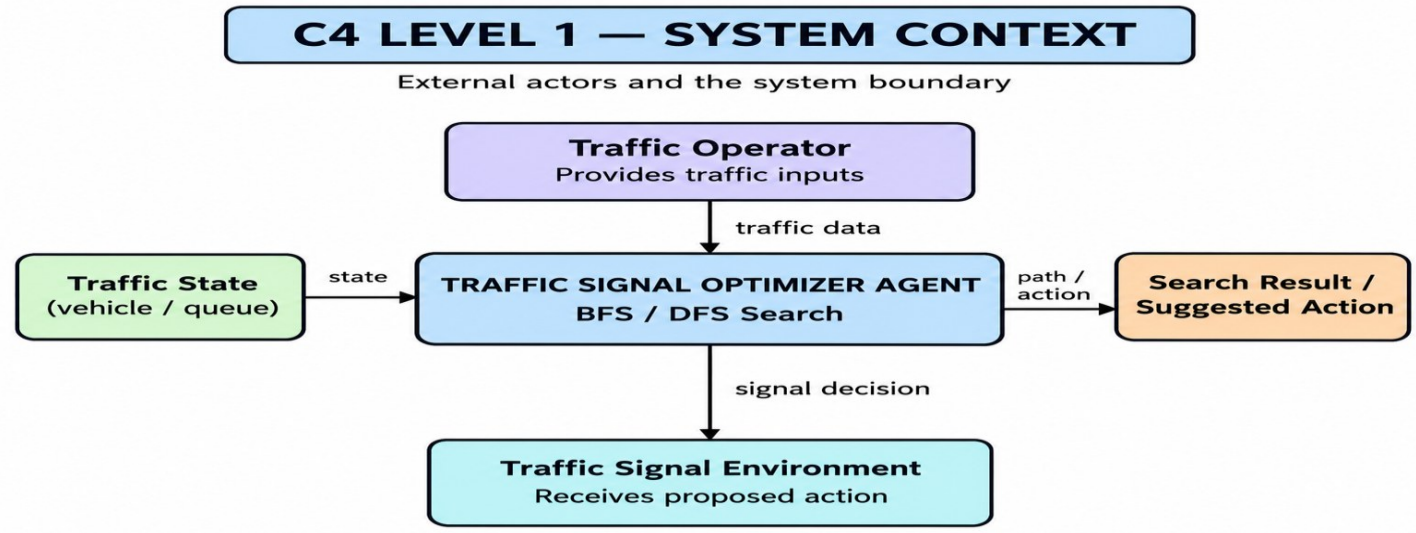


Figure 1: Context diagram of Traffic Signal Optimizer Agent.

The Traffic Operator provides or monitors traffic information. The Traffic Signal Optimizer Agent processes the traffic-state graph and runs BFS/DFS search. The selected signal action is passed to the Traffic Signal Controller.

3. Container Diagram (C4 Level 2)

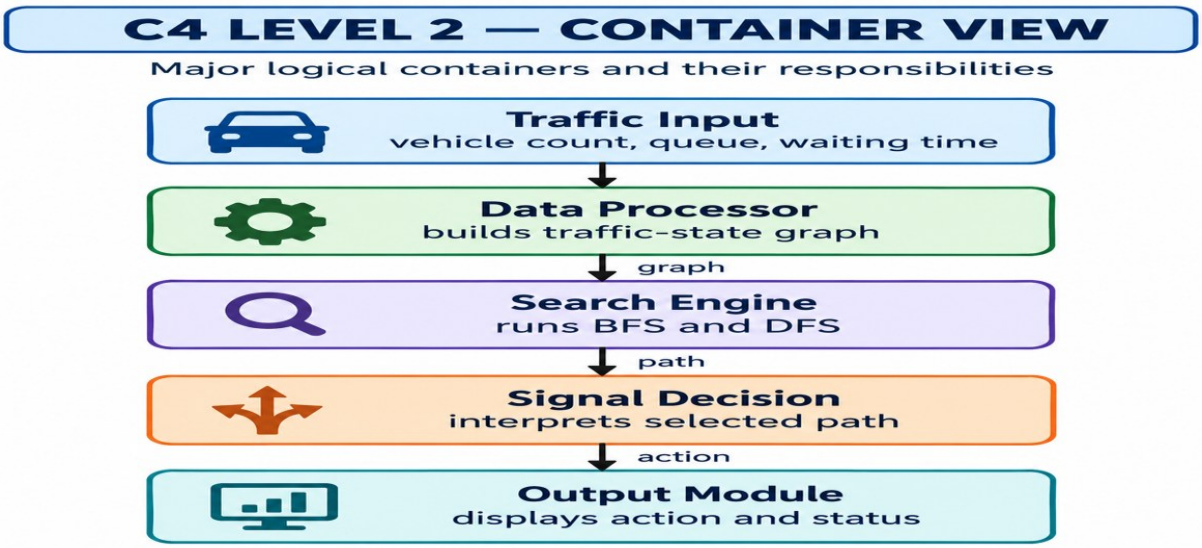


Figure 2: Container diagram.

Traffic Input – accepts vehicle count, queue length, waiting time and current signal state.

Data Processor – converts input into structured traffic states and prepares the graph.

Search Engine – runs BFS and DFS over the traffic-state graph.

Signal Decision – interprets the search result and maps the selected state/path to a signal action.

Output – displays the selected action and current traffic status.

4. Component Diagram (C4 Level 3)

Selected container: Search Engine

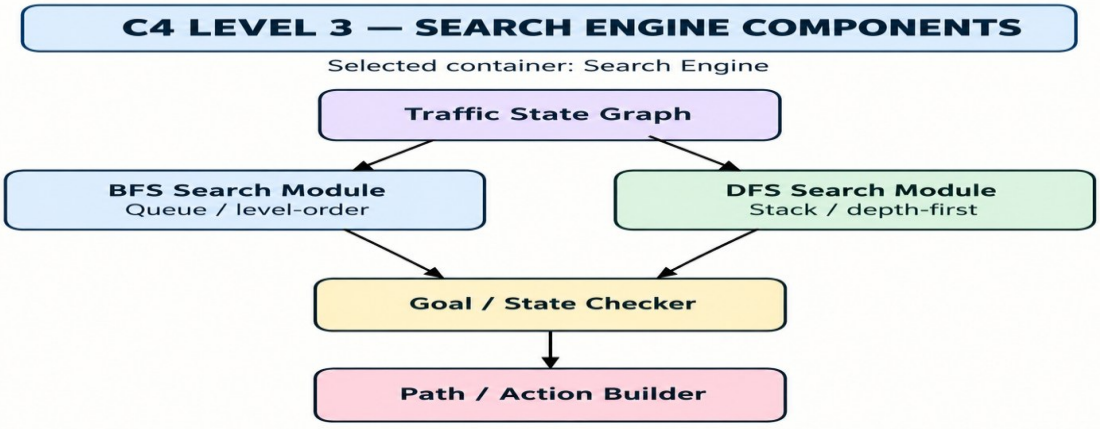


Figure 3: Component diagram of the Search Engine container.

Traffic State Graph – represents traffic conditions as nodes and possible signal changes as edges.

BFS Search Module – explores the graph level by level and can find a path to a goal state in an unweighted graph.

DFS Search Module – explores one branch deeply before backtracking and searching another branch.

Goal / State Checker – checks whether the desired traffic condition has been reached.

Path / Action – converts the resulting search path into a proposed next signal decision.

5. Code Level Overview (C4 Level 4)

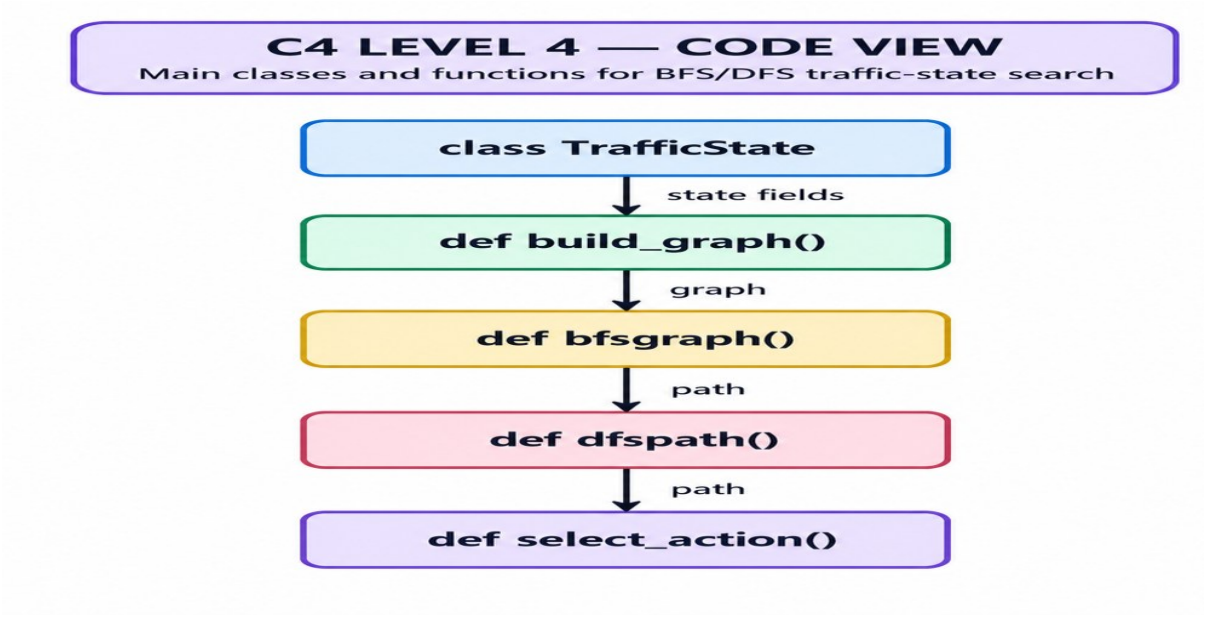


Figure 4: Code level overview.

class TrafficState – Stores vehicle count, queue length, waiting time and signal state.

def build_graph() – Creates traffic-state nodes and valid transitions.

def bfs() – Performs Breadth-First Search on the traffic-state graph.

def dfs() – Performs Depth-First Search on the traffic-state graph.

def select_action() – Uses the selected search result to propose a signal action.

6. Connection with SLE-2 BFS/DFS Work

This SLE-3 architecture extends the BFS and DFS work completed in SLE-2. The earlier experiment compared BFS and DFS on a graph from start node A to goal node AD using execution time and nodes

expanded. In this design, the same search algorithms are organized inside the Search Engine container of a Traffic Signal Optimizer Agent.

Traffic conditions are represented as graph states, and possible signal changes are represented as transitions. The SLE-2 measurements apply only to the earlier test graph and should not be interpreted as measured traffic-system performance.

7. Design Decisions

- Graph-based traffic representation: traffic conditions are represented as states so that BFS and DFS can search possible transitions.
- Separate BFS and DFS modules: the algorithms remain independent, making comparison and testing easier.
- Search Engine as the component focus: the core search logic is shown clearly at C4 Level 3.
- Modular architecture: input, processing, search, decision and output responsibilities are separated.
- Code-level overview: only the main class and function names are shown to keep the design readable.

8. AI Contribution Note

AI tool used: ChatGPT. AI assisted with organizing the system into the four C4 levels, drafting concise explanations, and preparing the diagram structure. The student should review the architecture, verify the report, and ensure that the proposed system matches the implementation and course requirements.

9. Conclusion

This SLE-3 report applies the Full C4 model to a Traffic Signal Optimizer Agent using BFS and DFS. The context, container, component and code-level views show how traffic input, graph preparation, search algorithms, signal decisions and output can be separated into clear responsibilities. The design connects the search techniques studied in SLE-2 with a proposed traffic-management application.